INCH-POUND

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w/ AMENDMENT 2
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SUPERSEDING
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w/ AMENDMENT 1
28 JULY 1991

MILITARY SPECIFICATION

POLYISOBUTYLENE

(FOR AMMUNITION USE)

This specification is approved for use by the U.S. Army Armament Research, Development and Engineering Center (ARDEC), and is available for use by all Department and Agencies of the Department of Defense

1. SCOPE

1.1 Scope. This specification covers one type of stabilized polyisobutylene of 99,000 to 120,000 average molecular weight for general use in Polyisobutylene Binder. This specification also covers another type of stabilized polyisobutylene of 99,000 to 210,000 average molecular weight for use in Polyisobutylene Binder of Composition C-4.

Comments, suggestions, or questions on this document should be addressed to: Commander, US Army ARDEC, Attn: RDAR-EIQ-SE, Picatinny Arsenal, New Jersey 07806-5000 or emailed to usarmy.picatinny.ardec.list.ardec-stdzn-branch@mail.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This section does not include documents cited in other sections of this standard or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the solicitation or contract.

FEDERAL STANDARDS

FED-STD-791 – Lubricants, Liquid Fuels, and Related Products, Methods of Testing

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-1916 – Department of Defense Preferred Methods for Acceptance of Product

(Copies of these documents are available online at http://quicksearch.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111-5094.)

2.3 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents are those cited in the solicitation or contract.

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM D460 – Standard Test Methods for Sampling and Chemical Analysis of Soaps, Iodine Number (Wijs) Method

ASTM D1524 – Standard Test Method for Visual Examination of Used Electrical Insulating Oils in the Field

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ASTM D3951 – Standard Practice for Commercial Packaging

(Copies of the above document can be obtained from the ASTM International headquarters, 100 Barr Harbor Drive, West Conshohocken, PA 19428 or online at www.ASTM.org .)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Material. The material use in the manufacture of polyisobutylene shall be such as to produce a finished polyisobutylene conforming to the requirements of this specification and all applicable specifications and drawings. The polyisobutylene shall be stabilized against depolymerization at 125° Fahrenheit and shall be in the form of consolidated crumbs to aid in incorporating the material in the binder composition.

3.2 Color. The color shall not be darker in color than the standard when determined as specified in 4.3.1 or the color value shall not be greater than 6½ when determined as specified in 4.3.1.1.

3.3 Chemical properties. The chemical properties of the polyisobutylene shall be in accordance with Table I.

TABLE I

Properties	Minimum	Maximum	Applicable Paragraphs
Intrinsic Viscosity	3.15	3.72*	4.3.2
Iodine Number	---	1.32	4.3.3
Chlorine, percent	---	0.10	4.3.4
Acidity (as HCl), percent	---	0.01	4.3.5
Insoluble matter, percent	---	0.20	4.3.6

*Maximum Intrinsic Viscosity value of 6.61 shall be used for use in Polyisobutylene Binder of Composition C-4

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4. QUALITY ASSURANCE PROVISIONS

4.1 General quality assurance provisions. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements specified herein. Except as otherwise specified, the supplier may utilize his own or any other inspection facilities and services acceptable to the Government. Inspection records of the examinations and tests shall be kept complete and available to the Government as specified in the contract or order. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements. Reference shall be made to Standard MIL-STD-1916 in order to define the terms used herein. Inspection shall be performed in accordance with this specification and other specifications referenced in any of the contractual documents.

4.1.1 Contractor quality assurance system. If the contractor desires to utilize a quality assurance system, which is at variance with the quality assurance provisions of 4.2 and 4.3 and other documents referenced herein, he shall submit a written description of the system to the contracting officer for approval prior to initiation of production. It shall include a description covering controls for lot formation and identification, inspection to be performed, inspection stations, sampling procedures, methods of inspection, (measuring and testing equipment,) and provisions for control and disposition of nonconforming material. The written description will be considered acceptable when, as a minimum, it provides the quality assurance provisions required by the provisions of 4.2 and 4.3 and the other documents referenced herein. The contractor shall not be restricted to the inspection station or the method of inspection listed in this specification provided that an equivalent control is included in the approved quality assurance procedure. In cases of dispute as to whether certain procedures of the contractor's system provide equal assurance, the comparable procedure of this specification shall apply. The contractor shall notify the Government of, and obtain approval for, any changes to the written procedure that effects the degree of assurance required by this specification or other documents referenced herein.

4.1.2 Submission of product. At the time the completed lot of product is submitted to the Government for acceptance the contractor shall supply the following information accompanied by a certificate which attests that the information provided is correct and applicable to the product submitted:

a. A statement that the lot complies with all quality assurance provisions of the approved current written description of the system.

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b. Quantity of product inspected.

c. Results obtained for all inspection performed shall be made available upon request by the Resident Ordnance Inspection.

d. Specification number and date, together with an identification and date of changes.

e. Certificates of analysis on all material procured directly by the contractor when such material is controlled by Government specifications listed in any of the contractual documents shall be made available upon request by the Resident Ordnance Inspection.

f. Quantity of product in the lot.

g. Date submitted.

The certificate shall be signed by a responsible agent of certifying organization. The initial certificate submitted shall be substantiated by evidence of the agent's authority to bind his principal. Substantiation of the agent's authority will not be required with subsequent certificated unless during the course of the contract, this authority is vested in another agent of the certifying organization.

4.1.3 Government verification. Using the contractor's written quality assurance procedure (see 4.1.1), this specification, and other contractual documents as a guide, the Government inspector shall verify all quality assurance operations performed by the contractor. Verification shall be in accordance with (a.) or (b.) below as applicable, the decision being the responsibility of the procuring activity. In either case, the inspector shall also ascertain, prior to acceptance, that all quality assurance provisions of other specifications referenced in any of the contractual documents have been complied with. Deviations from prescribed or agreed upon procedures discovered by the Government inspector shall be brought to the attention of the supplier. Disposition of the product and remedial action shall be as directed by the Government inspector and depending on the nature of the deviation, may consist of lot rejection, screening, re-sampling, re-instruction of the supplier's employees, or other appropriate action:

a. Verification at the point of manufacture shall be accomplished at unscheduled intervals in accordance with 4.1.3.1 and 4.1.

b. Verification at the point of delivery shall be in accordance with 4.1.

4.1.3.1 Surveillance. Surveillance shall include, but is not limited to:

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a. Observation of procedures concerning lot formation and identification.

b. Observation of sampling procedures and application of acceptance criteria.

c. Determination that all required examinations and test are performed in accordance with the prescribed procedures of this specification or approved equivalents thereto.

d. Review of procedures for control and disposition of nonconforming material.

4.2 Inspection provisions.

4.2.1 Lot formation. A lot shall consist of polyisobutylene from the same batch or blending operation from one manufacturer, in accordance with the same specification, or same specification revision, under one unchanged process. In event of a continuous process, a lot shall consist of material subjected to the same processing operations and condition.

4.2.2 Examination. Sampling plans and procedures for the following classifications of defects shall be in accordance with Standard MIL-STD-1916.

4.2.2.1 Container, sealed.

Category	Defect	Conformance Criteria	Paragraph Reference/Inspection Method	Code No. (see 6.3)
Critical	None defined			
Major				
101	Container damaged to the extent that the contents are exposed, or likely to become exposed or leak out, in subsequent handling	VL - IV	Visual	01001
Minor				
201	Marking misleading or unidentifiable	VL - II	Visual	01002

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4.2.3 Testing.

4.2.3.1 Sampling by lot. A random sample of 8 containers shall be selected from each lot. When lots are comprised of 8 containers or less each container shall be sampled.

4.2.3.2 Preparation of composite. Approximately two ounces of polyisobutylene shall be removed from each of the 8 containers by means of a sharp metallic instrument or other suitable tools in order to equal 16 ounces, and if there are less than ten containers, equal primary samples in sufficient quantity to equal 16 ounces, shall be removed from each container. The individual primary samples shall be combined in order to form a homogeneous composite sample of 16 ounces and subjected to the tests specified in 4.3. If the composite same fails to comply with any of the requirements specified, the lot shall be rejected.

4.3 Test methods and procedures.

4.3.1 Color - Code number 02001. A standard color solution shall be prepared by dissolving 0.0075 plus or minus 0.0001 gram of potassium dichromate in a small amount of distilled water in a one liter volumetric flask and diluted to volume with distilled water. A solution shall be prepared by dissolving 5.00 grams plus or minus 0.01 grams of the sample in 100 milliliters of diisobutylene. If the sample is not uniform in color, the darkest portion of the sample shall be used, in the preparation of the solution. A 100 milliliter portion of each of the above solutions shall be transferred to Nessler tube and placed beside each other on a white surface. The solution of the sample shall be no darker in color than the standard solution.

4.3.1.1 Alternate method. The color intensity can be determined by using a modified Hellige Aqua Tester, Model 611A. The color disks used in this instrument are standard disks known as Standard Varnish Color Disk #620 C-40. Colors on this disk are graduated from 1 to 9 and can be read to the nearest half color unit. Lower numbers indicate lighter color. Further details on test procedures can be found in ASTM method D1524-58T.

4.3.2 Intrinsic viscosity – Code No. 03001. One gram of the sample (cut into approximately 1 centimeter on a side) shall be dried in a vacuum oven at 70° Centigrade and reduced in pressure to 25 inches of mercury for a minimum of 24 hours. An accurately weighed portion of approximately 0.05 grams of the sample shall be transferred to a clean, dry, glass stoppered 50 milliliter cylinder. Fifty milliliters of diisobutylene (boiling point 101°C - 104°C.) shall be added to the cylinder by means of a calibrated pipette. The cylinder shall be tightly stoppered and placed in an end-over-end rotating device. The cylinder shall be allowed to rotate at approximately 2 revolutions per minute for 40 hours or as long as necessary to affect a complete solution at room temperature. The kinematic viscosity of the solution and of the solvent at 20° plus or minus 0.1° Centigrade shall be determined by means of a number 1

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Ubbelohde viscosimeter, in accordance with method 305 of FED-STD-791. The kinematic viscosity in centistokes from each of the above determinations shall be recorded. The intrinsic viscosity shall be calculated as follows:

$$\text{Intrinsic viscosity} = \frac{2.303}{C}\log_{10}\left(\frac{A}{B}\right)$$

A = viscosity of solution

B = viscosity of solvent

C = grams of polyisobutylene per 100 milliliters of diisobutylene

4.3.3 Iodine number – Code number 04001.

4.3.3.1 Reagents. The Wijs, potassium iodide and sodium thiosulphate solutions shall be made in accordance with ASTM method D460-58 iodine number Wijs method.

4.3.3.2 Procedure. An accurately weighed specimen of approximately 0.5 grams shall be transferred to a 500 milliliter Pyrex, glass stoppered, vapor-release iodination flask (see 6.2). Fifty grams of pure paradichlorobenzene shall be added to the flask and the flask placed on a hotplate with the vapor release of the flask lined up to release pressure. The solution shall be allowed to boil gently, and the flask and contents swirled from time to time. Care shall be taken to prevent particles of the sample from adhering to the sides of the flask above the solvent and scorching. When the sample has completely dissolved (approximately 20 to 30 minutes is usually required), the flask shall be removed from the hotplate and allowed to cool from room temperature. Fifty milliliters of chloroform shall be added before crystallization of the paradichlorobenzene is complete (i.e., when a small amount is still in the liquid phase). Five milliliters of the Wijs solution shall be transferred to the flask by means of a pipette and a thin film of the potassium iodide solution placed on the stopper, care being taken that none of the solution enters the flask. The flask shall be stopped with the vapor release closed, placed in an ice bath, and allowed to stand in a dark place for 2 hours. After removing the flask from the ice bath 25 milliliters of the potassium iodide solution and 50 milliliters of water shall be added. The excess iodide shall immediately be titrated with the sodium thiosulphate solution. Near the endpoint, 5 milliliters of a 1 percent starch indicator solution and 25 milliliters of ethyl alcohol shall be added to help break the emulsion. A blank determination shall be run as described above, using all the reagents. The iodine number shall be calculated as follows:

$$\text{Iodine number} = \frac{12.69(B-A)N}{W}$$

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Where:

A = milliliters of sodium thiosulphate used to titrate the sample.

B = milliliters of sodium thiosulphate used to titrate the blank.

N = normality of the sodium thiosulphate solution.

W = weight of sample.

4.3.4 Chlorine – Code number 05001.

4.3.4.1 Standard chloride solution. An amount of 1.649 grams of pure, dry sodium chloride shall be transferred to a 1-liter volumetric flask and diluted to the mark with water. The flask shall be well shaken to insure homogeneity of the solution. A 10 milliliter aliquot of the solution shall be transferred by means of a calibrated pipette, to a 100 milliliter volumetric flask and diluted with water to the mark. One milliliter of this solution is equivalent to 0.10 milligrams of chlorine. This solution shall be used to prepare standard solutions of chlorine.

4.3.4.2 Procedure. One gram of the sample cut into pieces shall be accurately weighed and transferred to a porcelain crucible. To the crucible shall be added 2.5 grams of sodium carbonate, and 2.5 grams of potassium nitrate and the ingredients mixed with the sample by means of a glass rod. The crucible and contents shall be covered with a porcelain cover and gentle fused over a Bunsen burner. The fusion shall be allowed to cool to room temperature. The crucible and contents shall be transferred to a 250 milliliter beaker, 35 milliliters of water added, and the contents heated to nearly boiling to dissolve the fused mass. The crucible shall be removed, the solution filtered, and the filtrate made slightly acid to litmus paper by the addition of concentrated nitric acid. The solution shall be transferred to a 50 milliliter Nessler tube, 5 milliliters of 10 percent silver nitrate added and diluted with water to the mark. The developed turbidity of the solution shall be compared to the standards. The standard shall be prepared by transferring aliquots of the standard chlorine solution described above to 50 milliliter Nessler tubes, 25 milliliters of water added and acidified with nitric acid. Five milliliters of 10 percent silver nitrate shall be added and diluted to the mark. The percentage of chlorine in the sample shall be calculated as follows:

Percent chlorine = $\frac{100A}{W}$

Where:

A = weight of chlorine found in sample by comparison by Standards.

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W = weight of sample.

4.3.5 Acidity – Code number 06001. Two grams of the sample cut into small pieces, approximately 1 centimeter on a side, shall be transferred to a 250 milliliter Erlenmeyer flask fitted with a ground glass stopper. 100 milliliters of xylene added, and the sample allowed to dissolve. The flask and contents may be placed on a mechanical agitator to aid solution of the sample. When the sample has completely dissolved, the solution shall be transferred to a separatory funnel, with 25 milliliters of distilled water and then shaken vigorously for about 30 seconds. The funnel shall be allowed to stand until the aqueous layer separates and then the aqueous layer shall be transferred to a 250 milliliter Erlenmeyer flask. The washings shall be repeated twice, with 25 milliliter portions of water, transferring each wash to the 250 milliliter Erlenmeyer flask. Two drops of phenolphthalein indicator shall be added to the aqueous solution and titrated with 0.05 normal sodium hydroxide solution to the first permanent pink appearance. A blank determination shall be conducted using the reagents and following the above procedure but omitting the sample. The acidity shall be calculated as follows:

$$\text{Percent acidity} = \frac{3.65(A-B)N}{W}$$

(As hydrochloric acid)

Where:

A = milliliters of sodium hydroxide to titrate sample.

B = milliliter of sodium hydroxide used to titrate the blank

N = normality of the sodium hydroxide solution.

W = weight of sample.

4.3.6 Insoluble matter – Code number 07001. Two grams of the sample cut into small pieces shall be accurately weighed and transferred to a 250 milliliter Erlenmeyer flask fitted with a ground stopper. One hundred milliliters of xylene shall be added and the sample allowed to dissolve. The flask and contents may be placed on a mechanical agitator to aid solution. When the sample has completely dissolved, it shall be filtered with the aid of a tared Gooch crucible. Washing shall be conducted with four 20 milliliter portions of xylene and one 20 milliliter portion of ethyl alcohol. If a white precipitate of polyisobutylene forms in the crucible upon addition of the alcohol, it shall be washed again with xylene, alcohol, and finally, ether. The crucible and contents shall be aspirated until no odor of ether remains. The crucible and contents shall be placed in an oven regulated to maintain a temperature of 100° to 105° Centigrade for one

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hour, cooled in a desiccator and weighed. The gain in weight of the Gooch crucible shall be calculated to percent insoluble matter.

5. PREPARATION FOR DELIVERY

5.1 Packing.

5.1.1 Level C. Unless otherwise specified, polyisobutylene shall be delivered in standard commercial containers, so constructed as to insure acceptance by common or other carrier for safe transportation, at the lowest rate, to the point of delivery.

5.2 Marking. In addition to any special marking required by the contract or order, shipment shall be marked in accordance with Specification ASTM D3951.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Ordering data. Procurement documents should specify the title, number and date of this specification.

6.2 A glass-stoppered flask as manufactured by the Scientific Glass Apparatus Company, Catalogue No. J-1728 has been found satisfactory.

6.3 Inspection code numbers. The five-digit code numbers assigned to the inspections herein are to facilitate future data collection and analysis by the Government.

6.4 Subject term (key word) listing.
Demolition
Composition C-4

6.5 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.

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